ELLIOT KAPLAN

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EMPLOYMENT

Stockholm University 2026 – Present

· Assistant professor

University of Mons 2025 – 2026

· FNRS postdoctoral fellow
· Visitor at the Hausdorff Center for Mathematics during October 2025

Max Planck Institute for Mathematics 2024 – 2025

· Postdoctoral fellow

McMaster University 2021 – 2024

· NSF postdoctoral fellow
· Postdoctoral fellow at the Fields Institute from July 2021 to June 2022
· Visitor at the Max Planck Institute for Mathematics during June 2023

EDUCATION

University of Illinois Urbana-Champaign

· Doctorate in mathematics, May 2021
· Dissertation: *Derivations on o-minimal fields*, advised by Lou van den Dries

Ohio University

· Bachelor of Science, Honors Tutorial College, May 2015
· Mathematics major, philosophy minor, graduated summa cum laude
· Undergraduate thesis: *Initial embeddings in the surreal number tree*, advised by Philip Ehrlich

RESEARCH INTERESTS

Model theory and its applications. Especially o-minimality, valued fields, and differential fields.

PUBLICATIONS

16. **Decomposing the automorphism group of the surreal numbers** 2026^+
With Lothar Sebastian Krapp and Michele Serra
Conference proceedings of DDG40 (to appear)

15. **Constant power maps on Hardy fields and transseries** 2026
J. Lond. Math. Soc., vol. 114, no. 3, paper no. e70709 (2026)

14. **Generic derivations, differential largeness, and $\mathbf{NTP}_2$** 2026
With Christoph Kesting
Ann. Pure Appl. Logic, vol. 177, no. 6, paper no. 103730, 9p. (2026)

13. **Monotone T-convex T-differential fields** 2025
With Nigel Pynn-Coates
Model Theory, vol. 4, no. 1, pp. 55–100 (2025)

12. **Hilbert polynomials for finitary matroids** 2024
With Antongiulio Fornasiero
Pacific J. Math., vol. 333, no. 2, pp. 273–308 (2024)

11. **A dichotomy for T-convex fields with a monomial group** 2024
With Christoph Kesting
MLQ Math. Log. Q., vol. 70, no. 1, pp. 99–110 (2024)

10. **Liouville closed H_T-fields** 2023
J. Algebra, vol. 628, pp. 265–327 (2023)

9. **T-convex T-differential fields and their immediate extensions** 2022
Pacific J. Math., vol. 320, no. 2, pp. 261–298 (2022)

8. **Surreal ordered exponential fields** 2021
With Philip Ehrlich
J. Symb. Log., vol. 86, no. 3, pp. 1066–1115 (2021)

7. **Generic derivations on o-minimal structures** 2021
With Antongiulio Fornasiero
J. Math. Log., vol. 21, no. 2, paper no. 2150007, 45p. (2021)

6. **Pairs of theories satisfying a Mordell-Lang condition** 2020
With Alexi Block Gorman and Philipp Hieronymi
Fund. Math., vol. 251, no. 2, pp. 131–160 (2020)

5. **Distality for the asymptotic couple of the field of logarithmic transseries** 2020
With Allen Gehret
Notre Dame J. Form. Log., vol. 61, no. 2, pp. 341–361 (2020)

4. **Continuous regular functions** 2020
With Alexi Block Gorman, Philipp Hieronymi, Ruoyu Meng, Erik Walsberg, Zihe Wang, Ziqin Xiong, and Hongru Yang
Log. Methods Comput. Sci., vol. 16, no. 1, paper no. 17, 24p. (2020)

3. **Logarithmic hyperseries** 2019
With Lou van den Dries and Joris van der Hoeven
Trans. Amer. Math. Soc., vol. 372, no. 7, pp. 5199–5241 (2019)

2. **Number systems with simplicity hierarchies II** 2018
With Philip Ehrlich
J. Symb. Log., vol. 83, no. 2, pp. 617–633 (2018)

1. **Resolution depth of positive braids** 2015
With David Krcatovich and Patricia O'Brien
J. Knot Theory Ramifications, vol. 24, no. 14, paper no. 1550075, 12p. (2015)

PREPRINTS

4. **Model completeness for the differential field of transseries with exponentiation** 2026
Preprint

3. **Building trees in large fields** 2026
With Angus Matthews and Erik Walsberg
Submitted

2. **Geometric fields, ranks, and generic derivations** 2026
With Antongiulio Fornasiero and Angus Matthews
Submitted

1. **Dimension theory for the asymptotic couple of the field of logarithmic transseries** 2025
With Allen Gehret and Nigel Pynn-Coates
Submitted

0. **Hyperserial fields** 2021
With Vincent Bagayoko and Joris van der Hoeven
Submitted

AWARDS AND FELLOWSHIPS

F.R.S.-FNRS Postdoctoral Research Fellowship (CR – Chargé de Recherches) 2025 – 2026

NSF Mathematical Sciences Postdoctoral Research Fellowship (MSPRF) 2021 – 2024

List of Teachers Ranked as Excellent by Their Students 2015, 2017 – 2020

Kuo-Tsai Chen Prize in Mathematics 2019

Waldemar J., Barbara G., and Juliette Alexandra Trjitzinsky Fellowship 2019

Graduate College Conference Travel Award 2016

U.S. Junior Oberwolfach Fellows NSF Grant (DMS-1641185) 2016

TALKS AT CONFERENCES, WORKSHOPS, AND SPECIAL SESSIONS

Beijing Model Theory Conference June 2026
Beijing, China

Differential Algebra and Related Topics XIV May 2026
Île d'Oléron, France

BeNeLux Mathematical Congress April 2026
Mathematical Logic Special Session.

Workshop on Tame Geometry and Combinatorics March 2026
University of Leeds, School of Maths

INdAM Intensive Period at the University of Naples May 2025
Mini-Workshop on Recent Developments in the Model Theory of Fields.

Centre International de Rencontres Mathématiques February 2025
Conference on Tame Geometry.

AMS-UMI International Joint Meeting in Palermo July 2024
Model Theory Special Session.

CMS Summer Meeting at the University of Saskatchewan June 2024
Mathematical Logic in Canada Special Session.

North American meeting of the Association of Symbolic Logic at Iowa State May 2024
Model Theory Special Session.

AMS Sectional Meeting at the University of Wisconsin-Milwaukee April 2024
Model Theory Special Session.

Banff International Research Station February 2024
Workshop on Exponential Fields.

Banff International Research Station February 2023
Workshop on Neostability.

Oberwolfach Research Institute for Mathematics November 2022
Mini-Workshop on Topological and Differential Expansions of o-minimal Structures.

Oberwolfach Research Institute for Mathematics December 2016
Mini-Workshop on Surreal Numbers, Surreal Analysis, Hahn Fields and Derivations.

Joint Math Meetings in Seattle, WA January 2016
AMS-ASL Special Session on Surreal Numbers.

SEMINAR AND COLLOQUIUM TALKS

Model Theory Seminar at the Kurt Gödel Research Center June 2026

PKU Model Theory Seminar March 2026

Oberseminar Algebra und Geometrie at TU Dortmund June 2025

Géométrie et Théorie des Modèles at the IHP May 2025

Séminaire de Logique, Algèbre, Arithmétique et Géométrie at UMONS April 2025

Logic Colloquium at the Kurt Gödel Research Center April 2025

Oberseminar Logik at Universität Bonn January 2025

AI Seminar at the Czech Technical University in Prague January 2025

Théorie des Modèles et Groupes at IMJ-PRG November 2024

Séminaire Général de Logique at IMJ-PRG November 2024

Logic Seminar at The Ohio State University April 2024

Logic Colloquium at the Kurt Gödel Research Center March 2024

Waterloo Logic Seminar January 2024

Oberseminar Modelltheorie, Geometrie und Gruppentheorie at Uni Münster June 2023

Guest talk at HHU Düsseldorf June 2023

Oberseminar Logik at Universität Bonn June 2023

Leeds Logic Seminar May 2023

Online Logic Seminar March 2023

Purdue University Model Theory and Applications Seminar March 2023

Logic Seminar at The Ohio State University February 2023

Waterloo Logic Seminar January 2023

Geometry and Model Theory Seminar at the Fields Institute October 2022

Fields Postdoc Colloquium April 2022

Geometry and Model Theory Seminar at the Fields Institute January 2022

Kolchin Seminar in Differential Algebra October 2021

Model Theory Seminar at the Fields Institute August 2021

Topological and Differential Expansions of O-minimal Structures Seminar February 2021

Manchester Logic Seminar June 2020

Notre Dame Logic Seminar June 2020

Séminaire de Logique Lyon-Paris May 2020

Caltech-UCLA Logic Seminar February 2020

Logic Seminar at The Ohio State University November 2019

Graduate Model Theory Seminar at UCLA January 2018

OUTREACH/GENERAL AUDIENCE TALKS

Graduate Colloquium at the University of Bonn June 2025
An introduction to model theory for a graduate student audience.

Casual Math Seminar at McMaster March 2023
An introduction to combinatorial game theory for an undergraduate/graduate student audience.

Undergraduate Friday Seminar at UIUC March 2021
An introduction to Conway's surreal numbers for an undergraduate student audience.

Summer Illinois Mathematics Camp at UIUC July 2018
A discussion of the basics of knot theory for a high school audience.

TEACHING AND MENTORSHIP

McMaster University 2022 – 2023

- Instructor for Linear Algebra I (Fall 2023).
- Instructor for Engineering Mathematics I (Fall 2022).

University of Illinois at Urbana-Champaign 2015 – 2020

- Discussion section leader for Calculus II (Fall 2015, Summer 2020), Calculus III (Fall 2016, Spring 2017, Fall 2018, Fall 2019), and Applied Linear Algebra (Spring 2018, Fall 2020).
- Head teaching assistant for Elementary Linear Algebra (Fall 2017) and Applied Linear Algebra (Spring 2018, Fall 2020).
- Grader for Mathematical Logic (Fall 2016, Fall 2017) and General Topology (Fall 2017).

Preliminary Arizona Winter School (PAWS) Fall 2022
The Southwest Center for Arithmetic Geometry
Course assistant for Ronnie Nagloo's course *Introduction to model theory with applications.*

Program for Interdisciplinary and Industrial Internships at Illinois Summer 2019
University of Illinois at Urbana-Champaign
Workshop leader for a three-day workshop on advanced linear algebra topics for graduate students.

Illinois Geometry Lab Summer 2019
Graduate Team Leader for a group of three undergraduate students. I designed and ran a research project on three-player combinatorial games.

Illinois Geometry Lab Spring 2018
Graduate Mentor for a group of four undergraduate students. I helped lead a research project studying functions recognizable by Büchi automata.

PROFESSIONAL SERVICE AND OUTREACH

Co-organizer: Oberseminar Logik at Universität Bonn — 2024 – 2025

Volunteer: OSU Math Department Booth at the COSI Science Festival — May 2024

Co-organizer: McMaster-Waterloo Model Theory Learning Seminar — Spring 2024

Co-organizer: ASL Special Session on Tame Geometry and Applications to Analysis — January 2023
At the Joint Math Meetings in Boston, MA

Co-organizer: Graduate Student Conference in Logic XXII — March 2021

Organizer: Nonstandard Analysis Learning Seminar at UIUC — Spring 2021

Graduate student member: Undergraduate Affairs Committee at UIUC — 2019 – 2020

Volunteer: Illinois Geometry Lab Booth at Illini Fest – Chicago — July 2019

Co-organizer: Graduate Student Conference in Logic XVIII — April 2017

OTHER WORK EXPERIENCE

Summer Intern — Summer 2016
Wolfram Research

Summer Researcher at UIUC — Summer 2015
Part of the Program for Interdisciplinary and Industrial Internships at Illinois

Undergraduate Researcher at Michigan State University — Summer 2014
Part of the Summer Undergraduate Research Institute in Experimental Mathematics

CRLA Certified Mathematics Tutor with Ohio University Tutoring Services — 2012 – 2015
Lead Mathematics Tutor from Fall 2014 through Spring 2015